Key Words: Buckling, buckling factor, plate

References

[1] S. Timoshenko, J.M. Gere: Theory of Elastic Stability. McGraw-Hill, New York and London, 1961

[2] A. Pflüger: Stabilitätsprobleme der Elastostatik. Springer Verlag, 1964

[3] K. Klöppel, D. Reuschling: Beulwerte von Rechteckplatten. Der Stahlbau 34 (1965), H. 11, S. 346-349

1 General

This HSB sheet provides buckling factors for rectangular plates with a certain range of aspect ratios for various loading types and support conditions. The buckling factors are applicable only to plates of isotropic material.

2 List of Symbols

Symbol	Unit	Description
a	mm	length of plate
b	mm	width of plate
k	–	buckling factor
k^*	–	modified buckling factor
t	mm	thickness of plate
E	MPa	Young's modulus
α	–	aspect ratio ($\alpha = a/b$)
β	–	stress ratio
ν	–	Poisson's ratio
σ, σ_x, σ_y	MPa	applied compressive stress
$\tilde{\sigma}$, $\tilde{\sigma_x}$	MPa	largest applied compressive stress in linear stress distribution (key value of stress)
σ_{cr}	MPa	critical buckling stress

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3 Analyses

Buckling is predicted to occur if the key value of the applied stress reaches the critical buckling stress, where the critical buckling stress is given by

$$\sigma_{cr} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1 - \nu^2)} \cdot \left(\frac{t}{b}\right)^2 . \tag{3-1}$$

With the modified buckling factor $k^* = k \cdot \dfrac{\pi^2}{12 \cdot (1 - \nu^2)}$ it follows

$$\sigma_{cr} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 . \tag{3-2}$$

For $\nu = 0.3$ the modified buckling factor is calculated as $k^* = k \cdot \dfrac{\pi^2}{12 \cdot (1 - 0.3^2)} = 0.904 \cdot k$.

It should be noted, however, that for most of the aluminum alloys used in airframe design $\nu = 0.33$, resulting in $k^* = k \cdot \dfrac{\pi^2}{12 \cdot (1 - 0.33^2)} = 0.923 \cdot k$.

In the Figs. 1 through 3 below, lower bound values or asymptotic values of the buckling factors k and k^*, respectively, are marked at the individual curves.

Fig. 1 shows the buckling factors for rectangular plates under a uniaxial stress state with linearly distributed stress versus the aspect ratio for a variety of support conditions.

Fig. 2 gives buckling factors for plates under a uniaxial stress state with uniformly distributed stress versus the aspect ratio also for a variety of support conditions.

Fig. 3 presents the buckling factors for rectangular plates under a biaxial stress state with uniformly distributed stresses where the plate is simply supported at all four edges. Buckling factor curves are shown for a variety of stress ratios σ_y/σ_x.

Figs. 4 through 7 provide the buckling factors for plates under biaxial loading, where the applied stresses are linearly distributed in x-direction and uniformly distributed in y-direction. The plates are simply supported at all four edges. The different figures are valid for different stress ratios of the stress in x-direction.

4 Application

The data provided for the critical buckling stress are valid for linear elastic conditions. The Young's modulus E is to be reduced if the plastic regime is entered.

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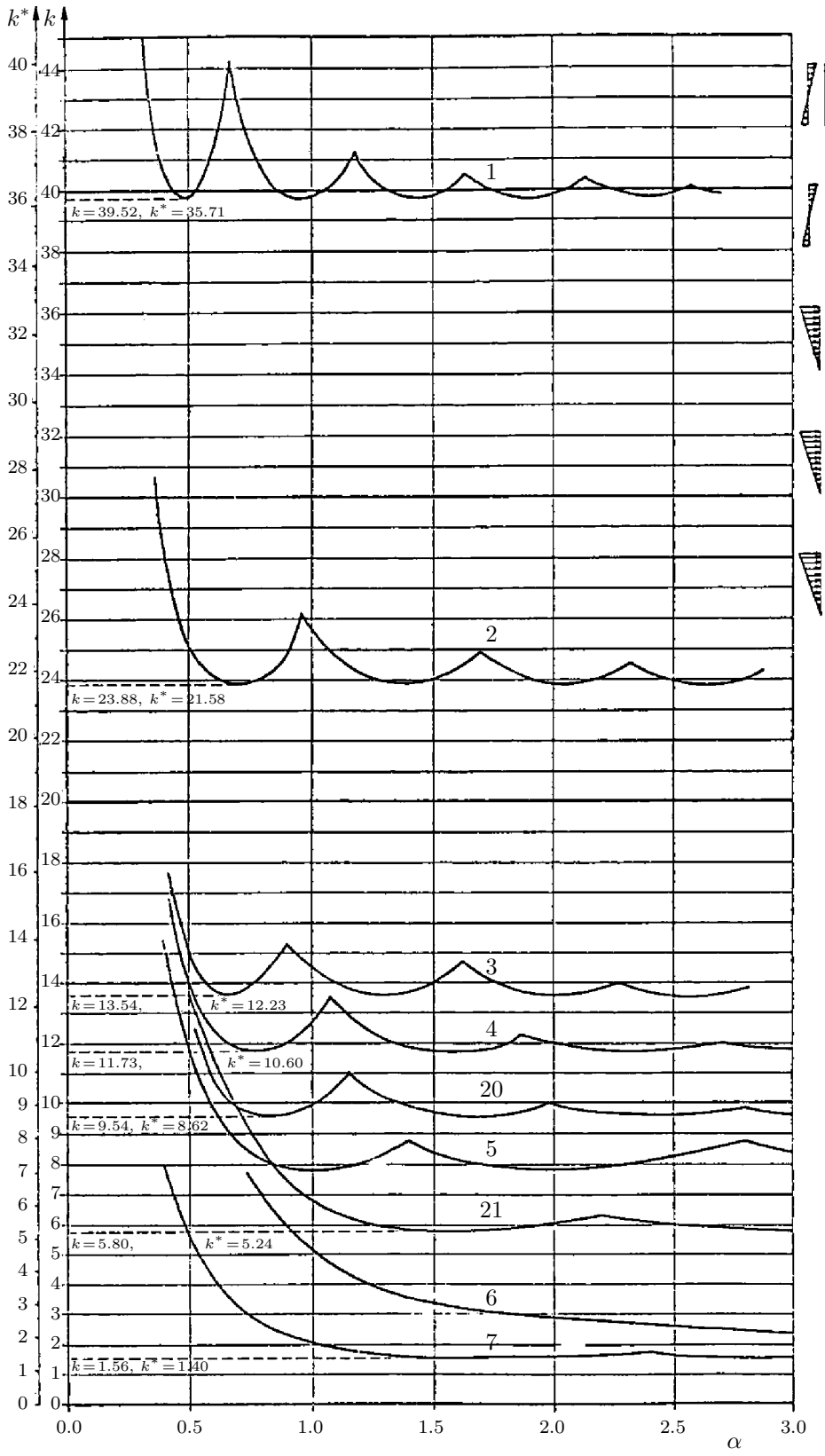


Figure 1: Buckling factors k and k^* (with $\nu = 0.3$) for a uniaxial stress state with linearly distributed stress (key value of stress is $\tilde{\sigma}$)

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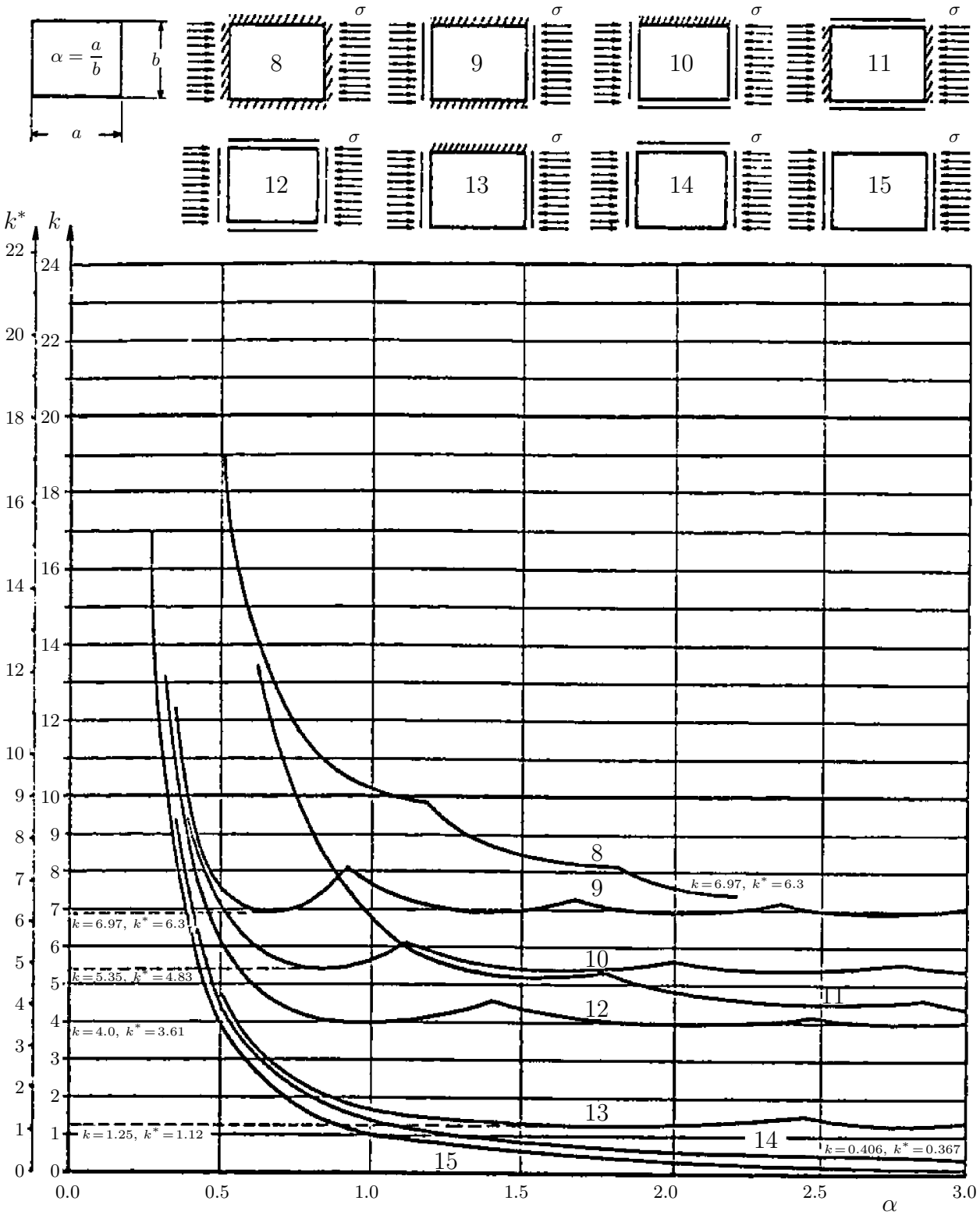


Figure 2: Buckling factors k and k^* (with $\nu = 0.3$) for a uniaxial stress state with uniformly distributed stress

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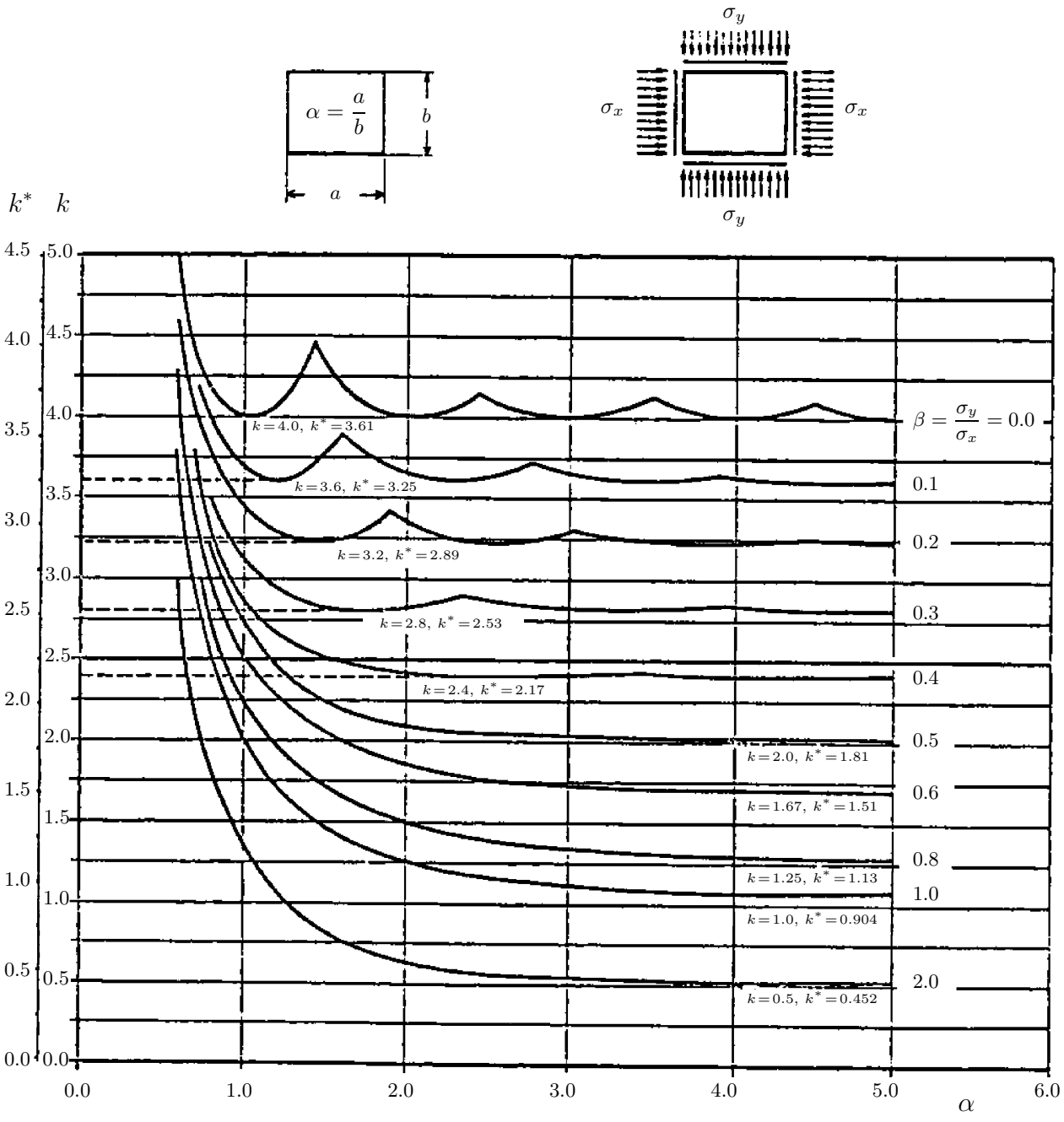


Figure 3: Buckling factors k and k^* (with $\nu = 0.3$) for a biaxial stress state with uniformly distributed stresses (key value of stress is σ_x)

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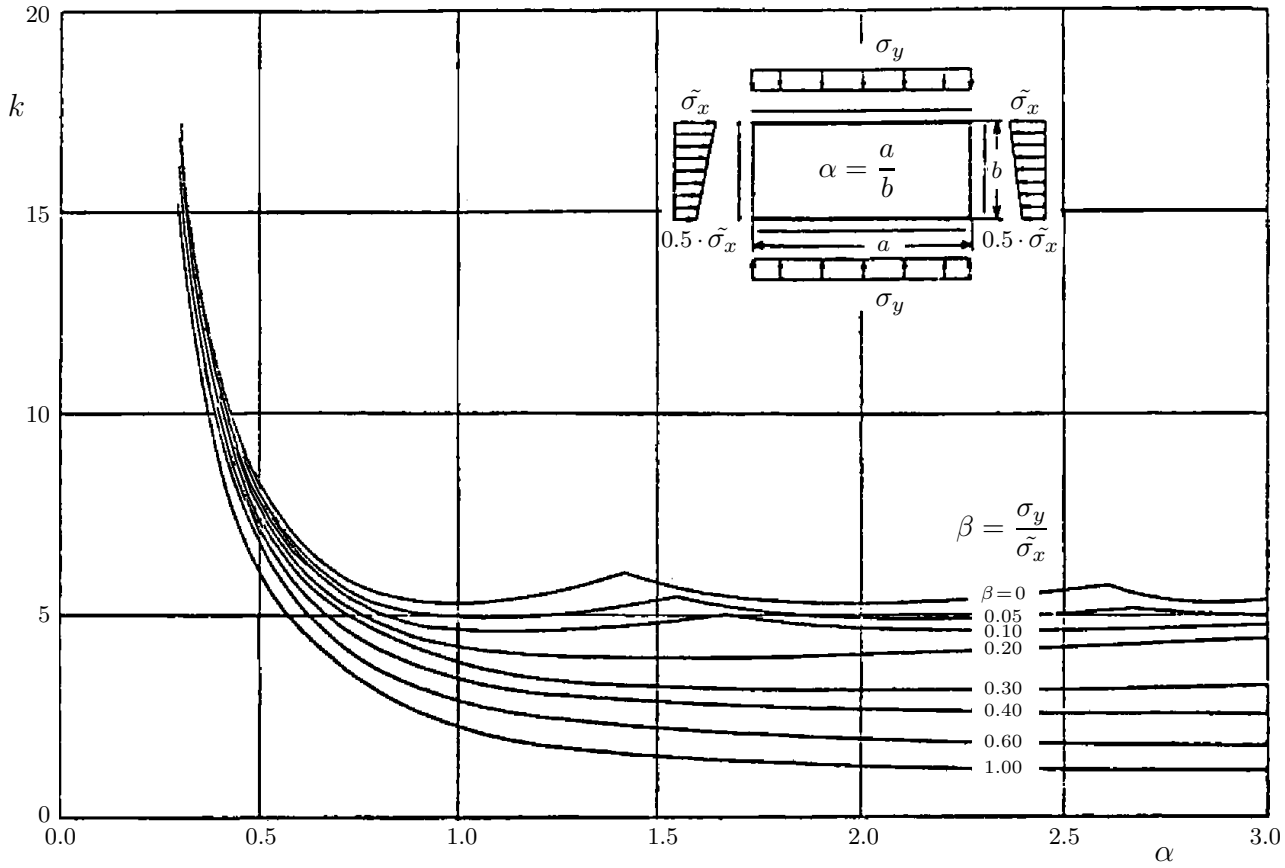


Figure 4: Buckling factor k for a biaxial stress state with linearly distributed stress in x-direction with a stress ratio of 0.5 and uniformly distributed stress in y-direction (key value of stress is $\tilde{\sigma_x}$)

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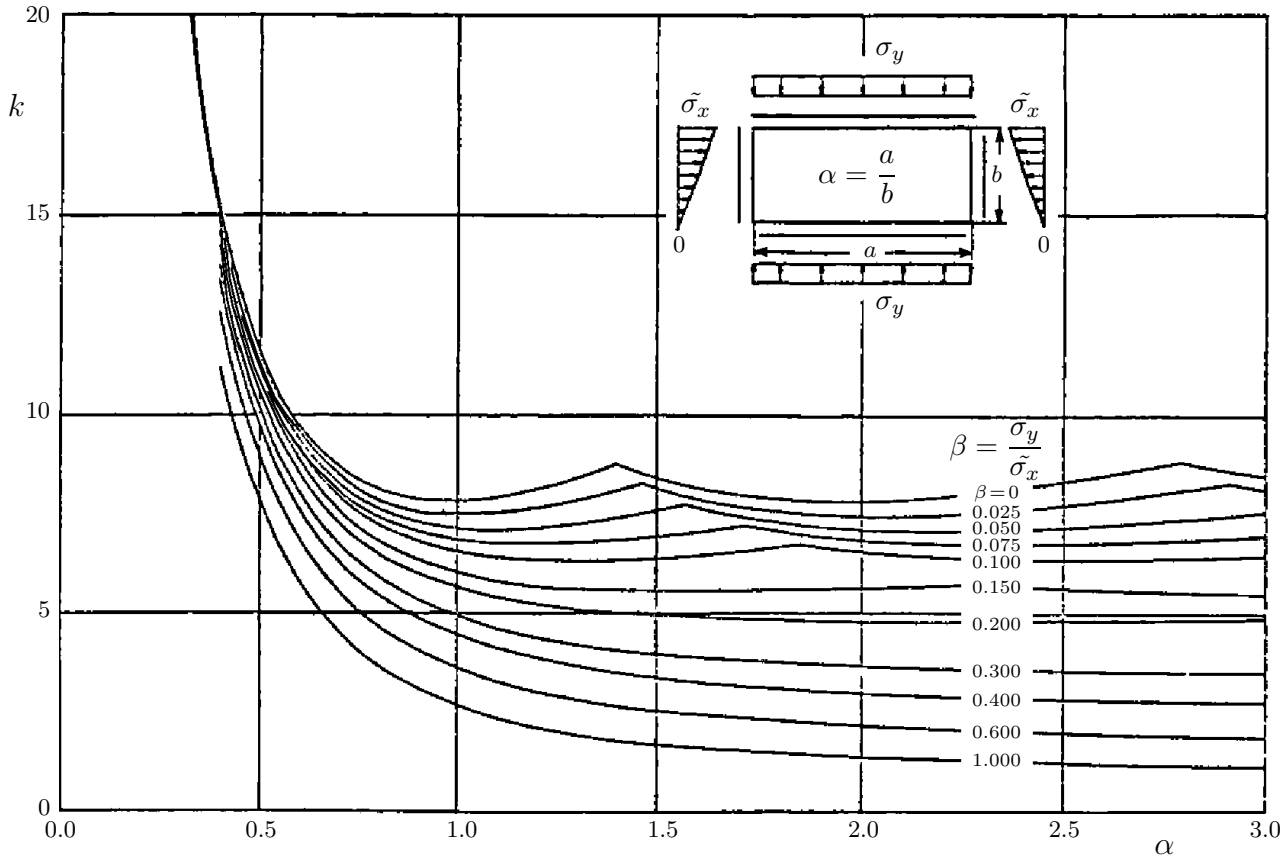


Figure 5: Buckling factor k for a biaxial stress state with linearly distributed stress in x-direction with a stress ratio of 0 and uniformly distributed stress in y-direction (key value of stress is $\tilde{\sigma_x}$)

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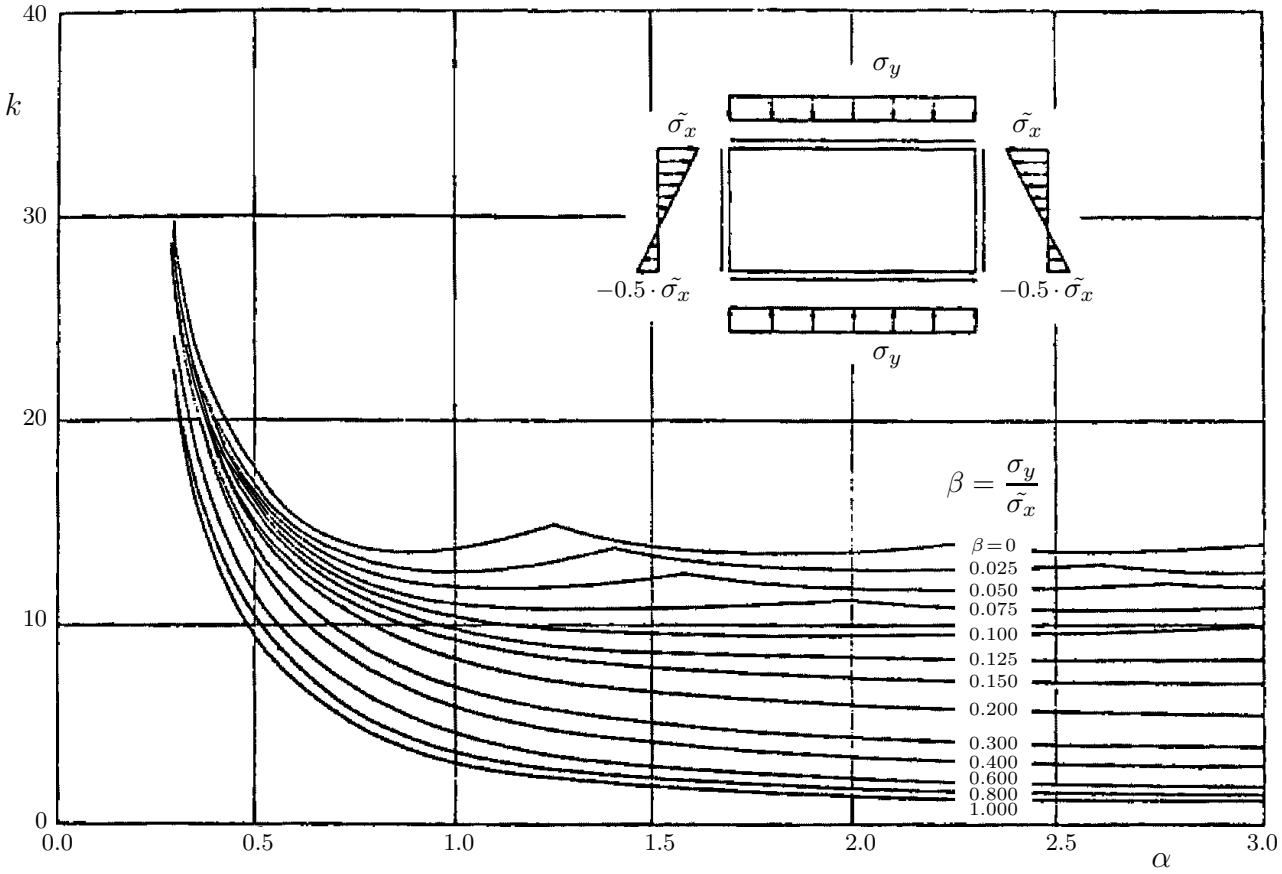


Figure 6: Buckling factor k for a biaxial stress state with linearly distributed stress in x-direction with a stress ratio of -0.5 and uniformly distributed stress in y-direction (key value of stress is $\tilde{\sigma_x}$)

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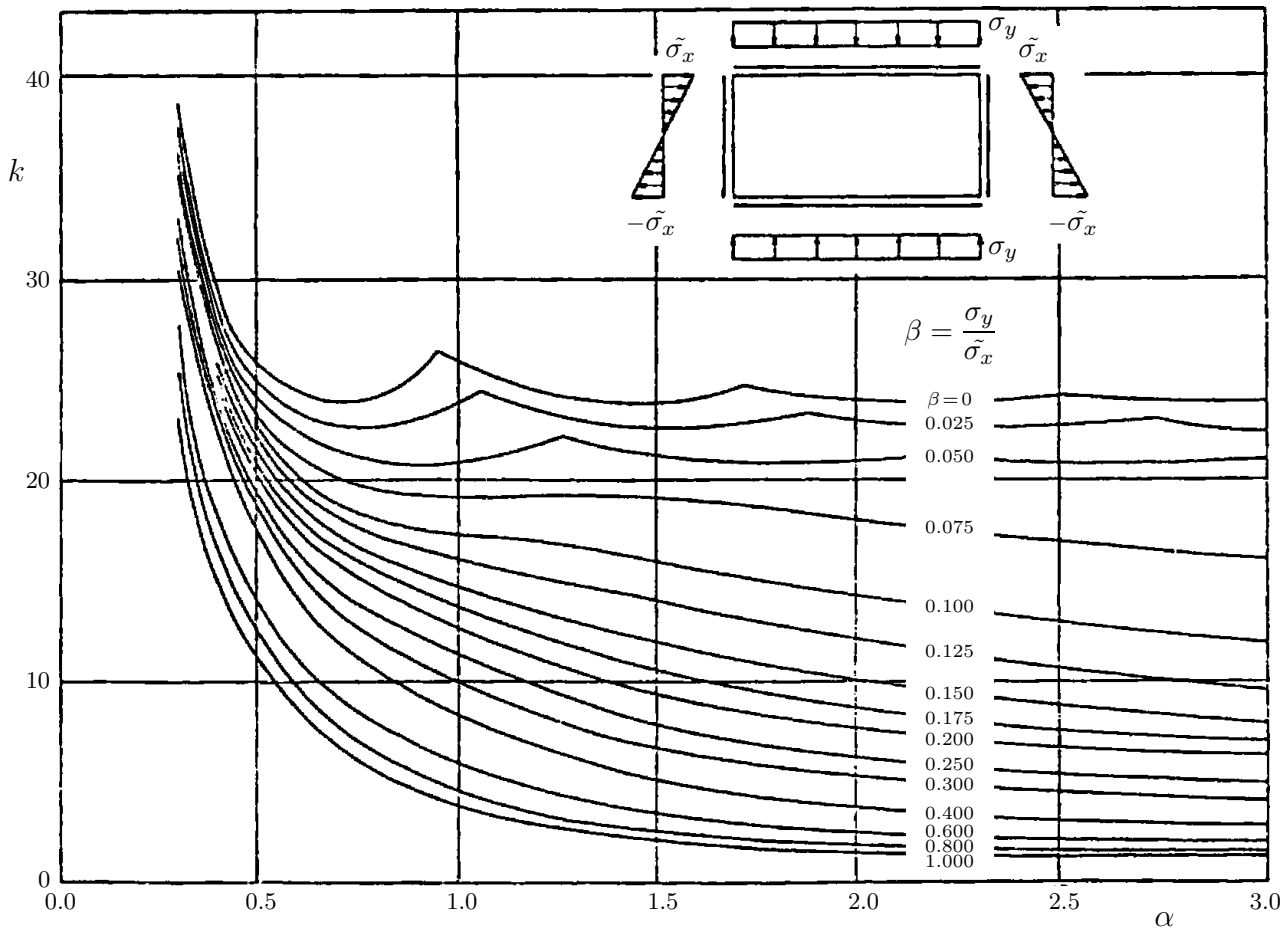


Figure 7: Buckling factor k for a biaxial stress state with linearly distributed stress in x-direction with a stress ratio of -1.0 and uniformly distributed stress in y-direction (key value of stress is $\tilde{\sigma_x}$)

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Key Words: Buckling, curved plate strip, cylindrical shell, compression loading

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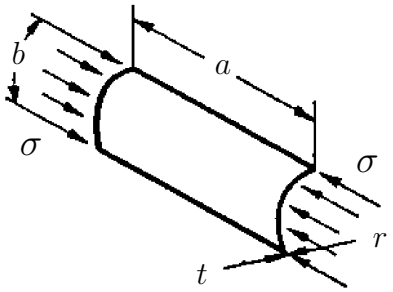
[1] N.N.: RAC - Structures Manual (Fig. 1.1712-1)

[2] N.N.: NACA-TN 1928

[3] N.N.: NACA-TN 3783

1 General

A method is presented to predict the critical buckling stress of curved plate strips under compression loading. The following configuration is considered.



It is a prerequisite that $a > b$. Data are provided for the support conditions 'simply supported' and 'clamped' at all four edges.

2 List of Symbols

Symbol	Unit	Description
a	mm	length of the plate strip
b	mm	arc width of the plate strip
k	–	buckling factor
k^*	–	modified buckling factor
r	mm	curvature radius
t	mm	thickness of the plate strip
E	MPa	Young's modulus
ν	–	Poisson's ratio
σ	MPa	applied compressive stress
σ_{cr}	MPa	critical buckling stress under compression loading

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3 Analysis

The critical buckling stress is determined by

$$\sigma_{cr} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1-\nu^2)} \cdot \left(\frac{t}{b}\right)^2 \tag{3-1}$$

$$= k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 \tag{3-2}$$

where

$$k^* = \frac{\pi^2}{12 \cdot (1-\nu^2)} \cdot k \,. \tag{3-3}$$

For the special case $\nu = 0.3$ one obtains the modified buckling factor $k^* = 0.904 \cdot k$. Buckling factor k and modified buckling factor k^* (for $\nu = 0.3$) are shown in Fig. 1.

Note however, that for most of the aluminum alloys used in airframe design $\nu = 0.33$, resulting in a slightly higher critical buckling stress.

4 Application

The data provided are valid as long as the critical buckling stress remains in the elastic regime. If the plastic regime is entered, the Young's modulus E is to be reduced accordingly.

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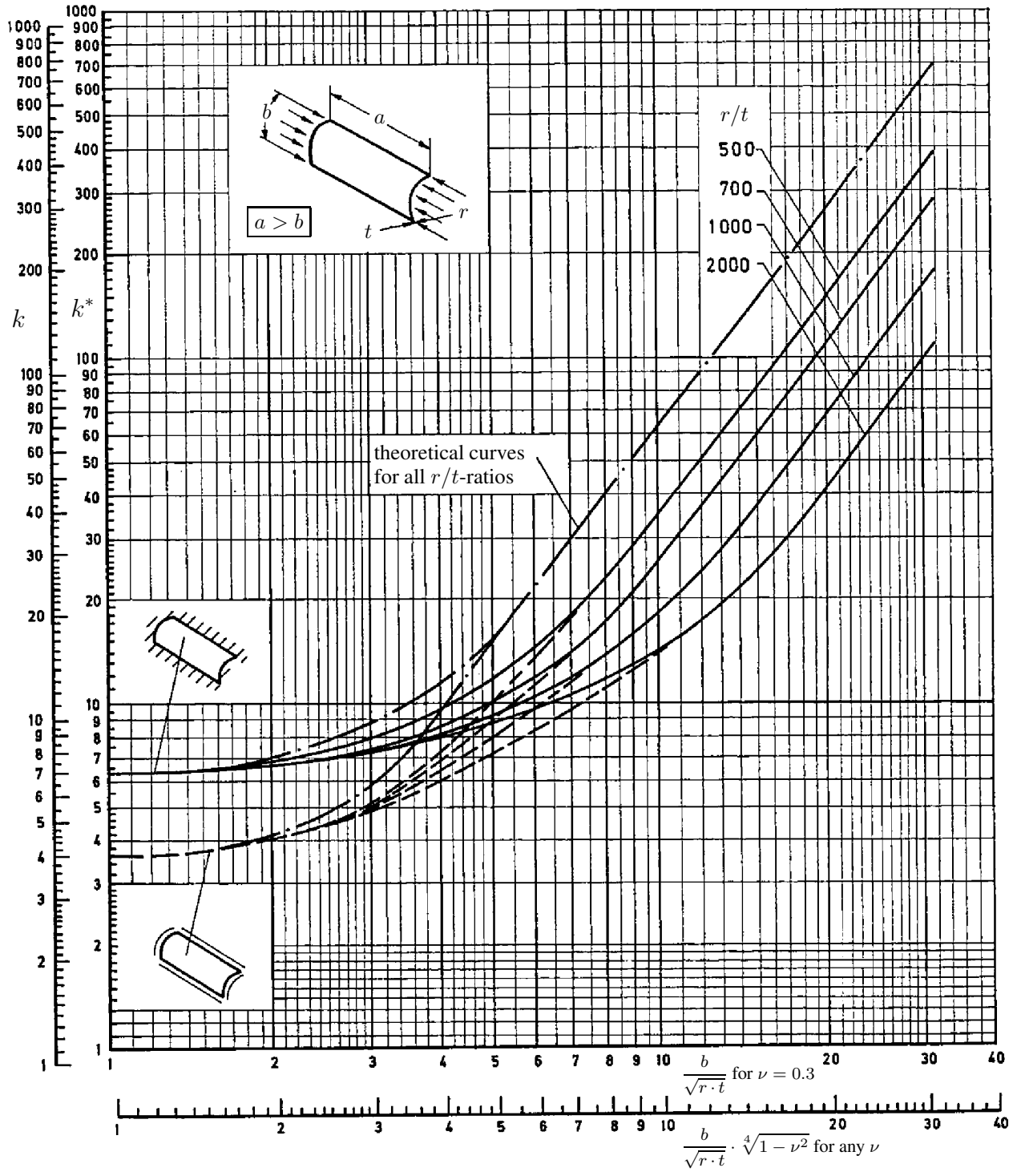


Figure 1: Buckling factor k vs. $\frac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1-\nu^2}$ for any value of ν and modified buckling factor k^* vs. $\frac{b}{\sqrt{r \cdot t}}$ for $\nu = 0.3$

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Key Words: Buckling, buckling factor, shear loading

References

[1] RAS, Data Sheet 02.03.01, 1963

1 General

This HSB sheet provides buckling factors for rectangular plates under shear loading for various support conditions and length/width ratios. The buckling factors are applicable only to plates of isotropic material.

2 List of Symbols

Symbol	Unit	Description
a	mm	length of plate
b	mm	width of plate
k	–	buckling factor
k^*	–	modified buckling factor
t	mm	thickness of plate
E	MPa	Young's modulus
ν	–	Poisson's ratio
τ_{cr}	MPa	critical buckling stress

3 Analyses

The critical buckling stress is given by:

$$\tau_{cr} = k \cdot \frac{\pi^2 \cdot E}{12 \cdot (1-\nu^2)} \cdot \left(\frac{t}{b}\right)^2 \tag{3-1}$$

With the modified buckling factor $k^* = k \cdot \dfrac{\pi^2}{12 \cdot (1-\nu^2)}$ it follows:

$$\tau_{cr} = k^* \cdot E \cdot \left(\frac{t}{b}\right)^2 \tag{3-2}$$

For $\nu = 0.3$ the modified buckling factor is calculated as $k^* = k \cdot \dfrac{\pi^2}{12 \cdot (1-0.3^2)} = 0.904 \cdot k$. It should be noted, however, that for most of the aluminum alloys used in airframe design $\nu = 0.33$, resulting in a slightly different critical buckling stress. The curves provided in Fig. 1 are valid for $\nu = 0.3$.

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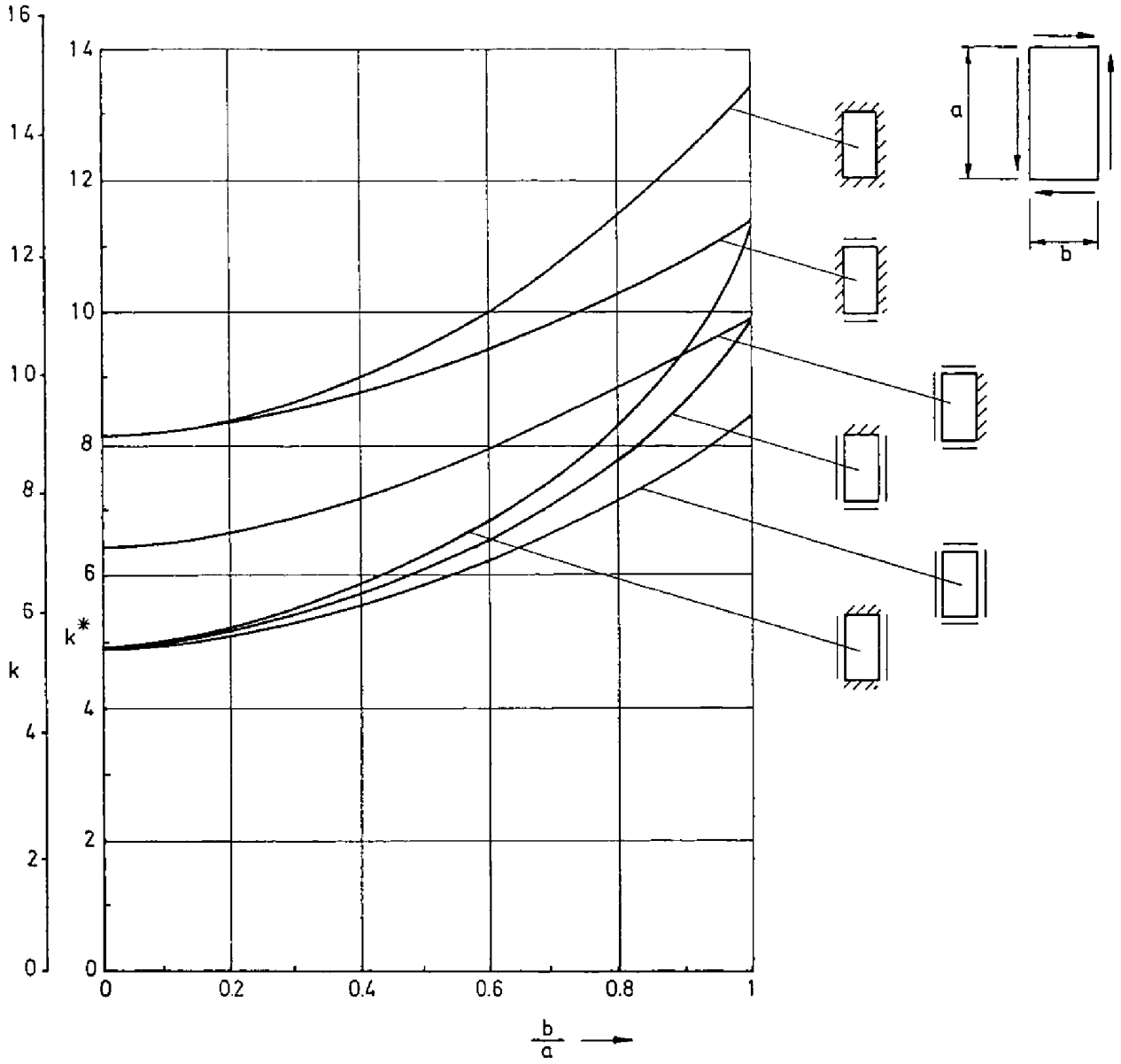


Figure 1: Buckling factor k and modified buckling factor k^* vs aspect ratio b/a

4 Application

The data provided for the critical buckling stress are valid for linear elastic conditions. The Young's modulus E is to be reduced if the plastic regime is entered.

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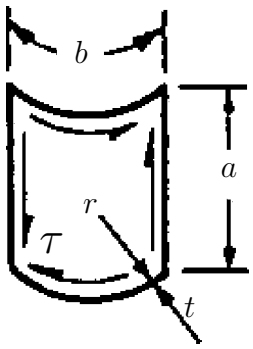
Key Words: Buckling, curved plate strip, cylindrical shell, shear loading

References

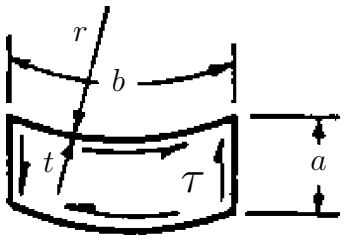
[1] N.N.: NACA-TN 3783

1 General

A method is presented to predict the critical buckling stress of curved plate strips under shear loading. The following two configurations are considered.



Configuration A
(long curved plate: $a \geq b$)



Configuration B
(wide curved plate: $b \geq a$)

Data are provided for the support conditions 'simply supported' and 'clamped' at all four edges.

2 List of Symbols

Symbol	Unit	Description
a	mm	length of the plate strip
b	mm	arc width of the plate strip
k_s	–	buckling factor in accordance with shear loading
k_s^*	–	modified buckling factor in accordance with shear loading
r	mm	curvature radius
t	mm	thickness of the plate strip
E	MPa	Young's modulus
ν	–	Poisson's ratio
τ	MPa	applied shear stress
τ_{cr}	MPa	critical buckling stress under shear loading

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3 Analysis

The critical buckling stress is determined by

$$\tau_{cr} = k_s \cdot \frac{\pi^2 \cdot E}{12 \cdot (1-\nu^2)} \cdot \left(\frac{t}{b}\right)^2 \tag{3-1}$$

$$= k_s^* \cdot E \cdot \left(\frac{t}{b}\right)^2 \tag{3-2}$$

where

$$k_s^* = \frac{\pi^2}{12 \cdot (1-\nu^2)} \cdot k_s \, . \tag{3-3}$$

For the special case $\nu = 0.3$ one obtains the modified buckling factor $k_s^* = 0.904 \cdot k_s$. Note however, that for most of the aluminum alloys used in airframe design $\nu = 0.33$, resulting in a slightly higher critical buckling stress.

Buckling factor k_s and modified buckling factor k_s^* (for $\nu = 0.3$) are shown in Figs. 1 through 4 for both the configurations A and B as well as for the support conditions 'simply supported' and 'clamped' at all four edges.

4 Application

The data provided are valid as long as the critical buckling stress remains in the elastic regime. If the plastic regime is entered, the Young's modulus E is to be reduced accordingly.

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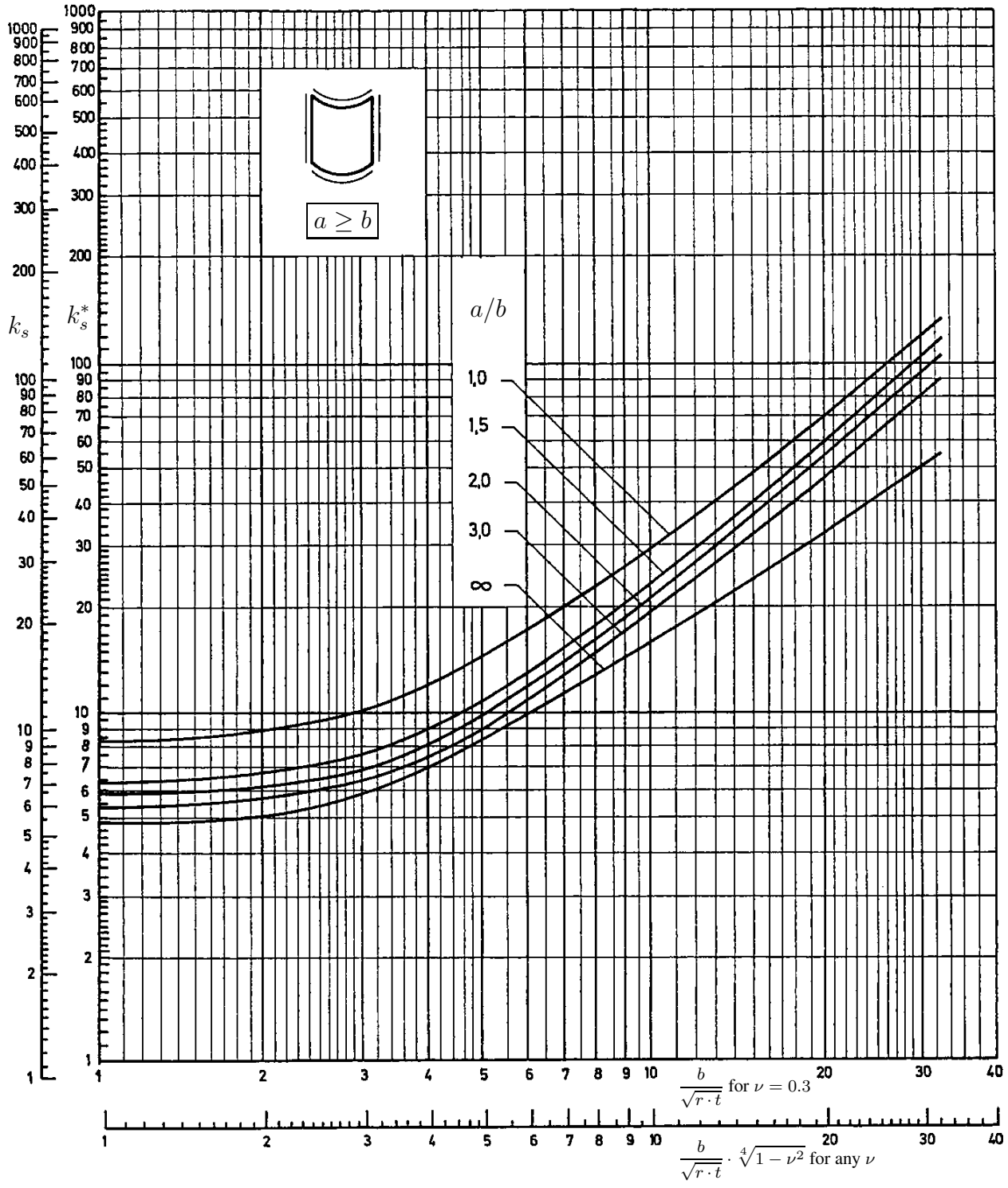


Figure 1: Buckling factor k_s vs. $\dfrac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1-\nu^2}$ for any value of ν and modified buckling factor k_s^* vs. $\dfrac{b}{\sqrt{r \cdot t}}$ for $\nu = 0.3$ (configuration A, support condition: simply supported)

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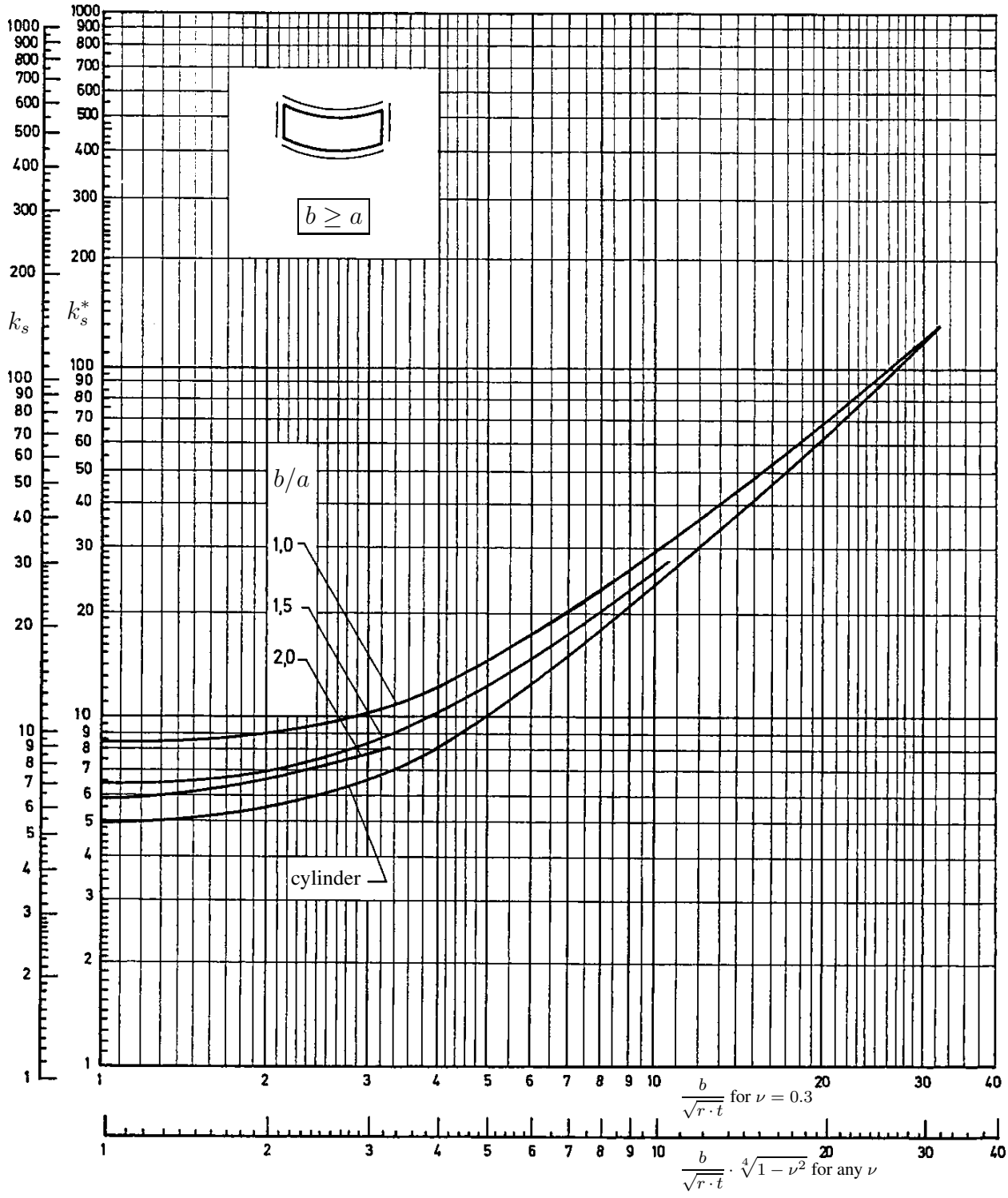


Figure 2: Buckling factor k_s vs. $\dfrac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1-\nu^2}$ for any value of ν and modified buckling factor k_s^* vs. $\dfrac{b}{\sqrt{r \cdot t}}$ for $\nu = 0.3$ (configuration B, support condition: simply supported)

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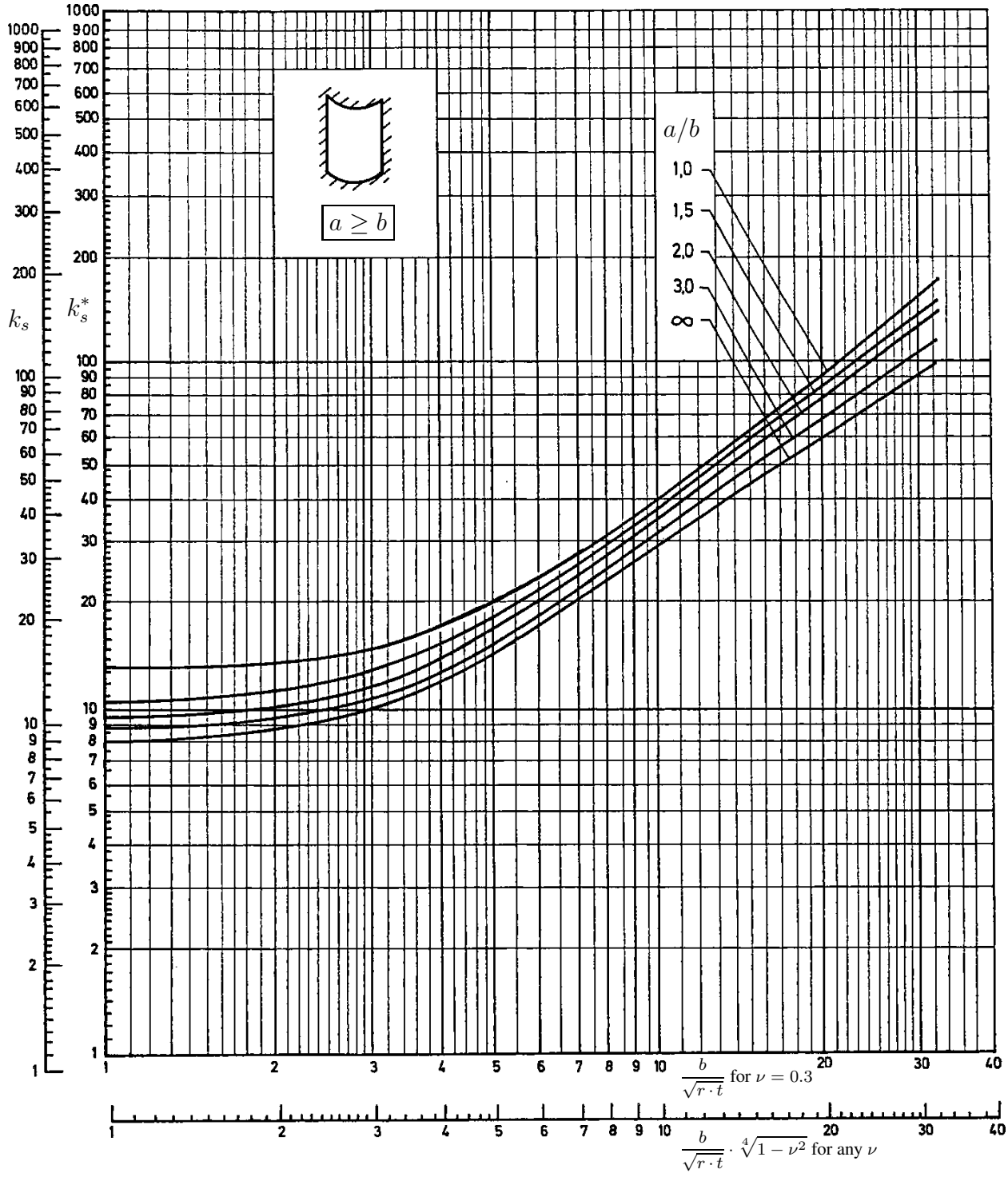


Figure 3: Buckling factor k_s vs. $\dfrac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1-\nu^2}$ for any value of ν and modified buckling factor k_s^* vs. $\dfrac{b}{\sqrt{r \cdot t}}$ for $\nu = 0.3$ (configuration A, support condition: clamped)

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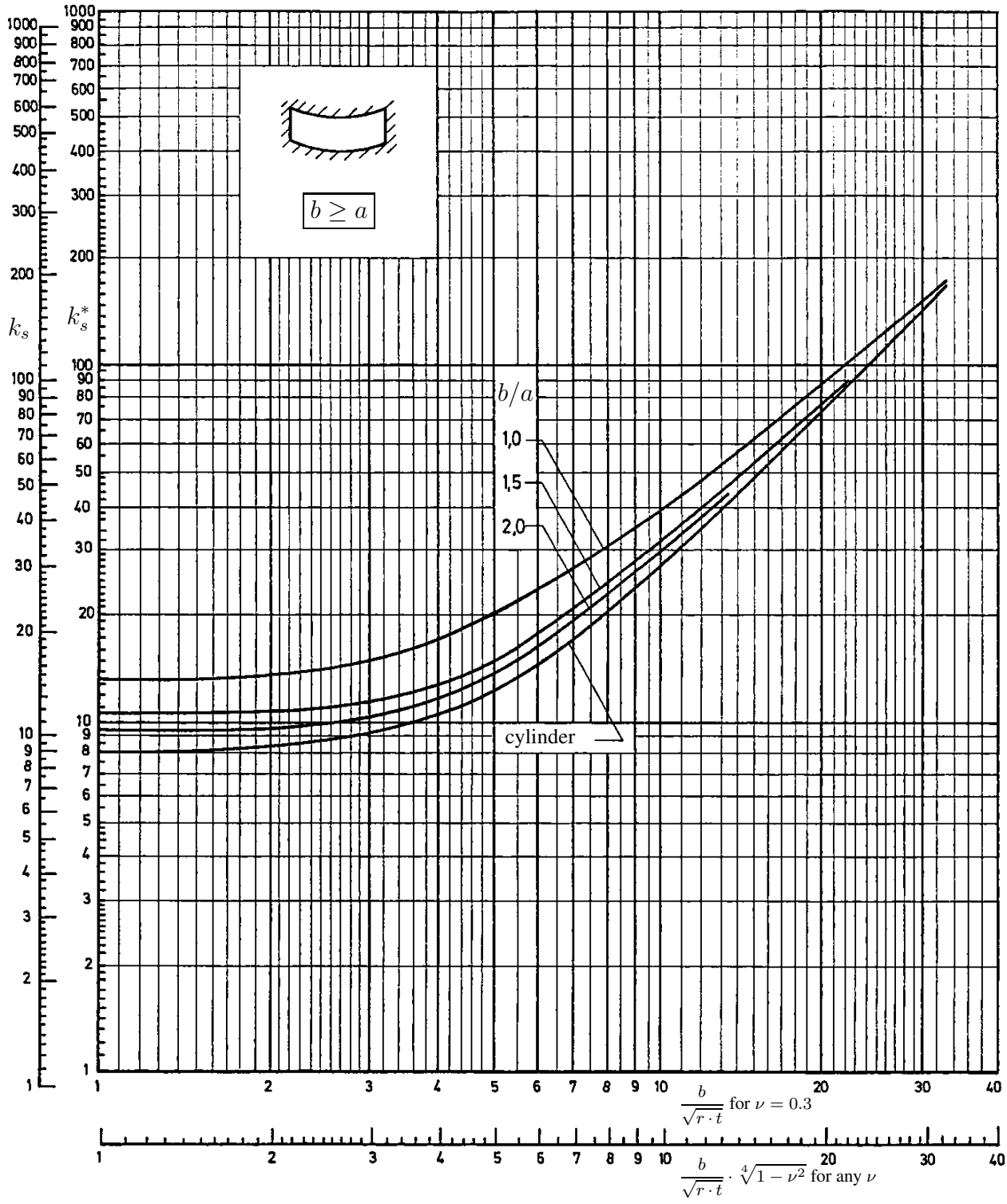


Figure 4: Buckling factor k_s vs. $\dfrac{b}{\sqrt{r \cdot t}} \cdot \sqrt[4]{1-\nu^2}$ for any value of ν and modified buckling factor k_s^* vs. $\dfrac{b}{\sqrt{r \cdot t}}$ for $\nu = 0.3$ (configuration B, support condition: clamped)

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